Handoff 23: The new README (README-01)

Date: 2026-10-27 (America/Toronto)
Written by: Claude Code (implementer), in a separate git worktree
For: the human maintainer, and ChatGPT or Claude chat (reviewer) without repository access
Status: **in_review**: one pull request (#28), documentation only, not merged. No code change. Polished after review (section 12).

> **How to use this file:** upload it with the new `README.md`, `DEPLOYMENT.md` and the three screenshots. The pull request diff is the source of truth (`build.md` section 14). Commit `9692b03` is the draft exactly as it came from chat; the next commit holds every correction, so the diff between them shows what changed.

Starter prompt for the chat assistant

```
You are reviewing Python Academy's new README (README-01). It was drafted
in chat, then checked line by line against the code, tests and docs,
corrected, and extended with every still-true fact from the old README.
Three screenshots and a banner were added. I'm attaching the README,
DEPLOYMENT.md and this handoff. Check that every claim is true and
not overstated (section 3 lists each claim with its source), that
nothing a reader needs was lost, and that the page reads well for a
beginner and for a developer. Explain at a first-year computer
engineering level.
```

1. Step 0 (checks before editing)

Check	Result
`git fetch --all`, `gh pr list --state all`	Done. One PR open: #27 (UI-05); this PR is not stacked on it
Earlier README-01 work	**None**
Where the work happened	A separate git worktree, `../python-academy-readme`, on a new branch `docs/readme` from `origin/main` at `3263342`. The main folder and the other worktrees were not switched or changed

Check	Result
The drafts	`README-draft-from-chat.md` and `README-banner-from-chat.svg` were copied in as `README.md` and `docs/images/banner.svg`, then **moved out of the main folder** to my session scratch folder (section 9). Neither is committed under its draft name

2. Handoff record (build.md template)

Field	Value
Task	README-01: publish the README drafted in chat, with every claim verified, nothing true lost, three screenshots and the banner
Human goal	Anyone opening the repository understands what works today, what does not, and how it is built, without being misled
Repository and branch	`torontojohn/python-academy` (private), branch `docs/readme`
Commits	`9692b03`: the draft and banner as received. `1217eb0`: corrections, kept facts, screenshots. Then this handoff. Then the polish (section 12)
Files changed	`README.md` (replaced), `DEPLOYMENT.md` (one section added), `docs/images/banner.svg`, `catalog.png`, `friendly-error.png`, `hint-ladder.png` (new), this handoff
Not changed	Any code, `docs/plan/backlog.md`, `docs/plan/decision-log.md`, the banner's design
Deployment status	Not requested; documentation only

3. Every claim, and where it was checked

Claim in the README	Source	Result
Live site and preview links	`DEPLOYMENT.md`; `curl`	Both answer 200
"Started as a way to help one grade 9 student…"	**Not in the repository.** `build.md` section 1 says the maintainer is the owner's son, a first-year engineering student; `site-design.md` writes for grade 9 beginners	**Kept as written; you must confirm** (question 1)
Khan Academy-style catalog growing into git, web, data, SQL, Node.js, AI	`backend/content/catalog.json`, `build.md`, `teaching-design.md` 6.6	True
Python Foundations unit 1 released, the rest "Coming soon"	`catalog.json`: `py` has 12 units, 1 released; 10 other courses are `coming_soon`	True

Claim in the README	Source	Result
The course titles	`catalog.json`	**Corrected:** "Web Foundations: HTML and CSS"; Hackathon Ready has 7 planned units
Catalog, course pages, sidebar, rule-based recommendation from a goal	`frontend/app/catalog.tsx`, `course.tsx`, `adventure.tsx`, `backend/app/recommend.py`	True
Five lessons, five cards each	Content files; `CARD_SEQUENCE` in `content.py`	True
Pyodide worker with a time limit, an output limit, no network	`python-worker.js` (20,000 characters, network APIs removed), `adventure.tsx` (15-second stop)	True; **the numbers were added**
Friendly errors, line marked, Python's message one click away, three-step hint ladder, no answer button, tagged hint first	`app/feedback.ts`, `code-editor.tsx`, content `error_hints`; handoffs 15 and 19	True
Result labels	`STATE_LABELS` in `app/feedback.ts`	**Corrected** to the words the page shows: "Check passed", "Not yet", "Python stopped", "Not checked", "Check unavailable", and "Ran · not checked"
XP and badges	`awards.py` (`CHALLENGE_XP = 100`), the award tests	True; **"100 XP and a badge, awarded once" added**
Parent and student accounts, invitations, parent progress view	`main.py` (`enroll`, `invite`, `family`)	True
Preview saves nothing	`adventure.tsx` (preview uses `/api/check`, not `/api/attempts`)	True
Near-term work order	`docs/plan/backlog.md` section 2 and the rows	True. Per-unit lessons are in PR #27, not on `main` yet, so "coming" is right
The diagram and its four bullets	`next.config.mjs` (rewrites), `proxy.ts`, `main.py`, `browser_checks.py`, `content.py`	True; renders with `mmdc` (section 6)

Claim in the README	Source	Result
Sign-in: Google OIDC with state and nonce, `openid email profile`, self-declared age gate (over 25, 13+)	`main.py`, `DEPLOYMENT.md`, `age_on` and `enroll`	True; "through Authlib" and "never a birthday scope" **kept from the old README**
Sessions: random ID, SHA-256 digest, logout ends it, 24 h or 7 days, admin 1 h or 8 h	`sessions.py`, `db.py` (`UserSession`), `DEPLOYMENT.md`	True; cookie flags (`HttpOnly`, `SameSite=Lax`, `Secure` in production) **added** from `main.py`
Rate limits on sign-in, checks and writes, per account or address	`limits.py` `RULES`; `test_every_write_endpoint_has_a_limit`	True; "including invitations" and "429 with `Retry-After`" added
Headers: HSTS, nosniff, no-referrer, framing blocked	`next.config.mjs`, `main.py`	True
Nonce CSP in Report-Only; one-line switch; worker policy enforced	`proxy.ts` (`ENFORCE = false`), `next.config.mjs`	True; "the worker policy only allows the Pyodide CDN, and learner code has no network functions" added
Admin from `ADMIN_EMAILS`, verified email, re-checked every request, 401 and 403	`admin.py`, `test_admin.py`	True; **"at sign-in" added** (being listed later only counts from the next sign-in)
Testers marked by the admin	`admin.py`, `frontend/app/admin/testers/page.tsx` (PR #26)	True; "on a page the API shows only to them" added
Feedback anonymous, 50 days, 90 items, button and inbox still to come	`backend/app/feedback.py`, `test_feedback.py`, backlog FB-05 and FB-06	True
No secrets in the repository	`.env.example` has empty values; `render.yaml` uses `sync: false` and generated values	True
Known limits	`browser-checks.md` section 5, `limits.py`, the README and handoffs	True; "backup and restore drills" and "security and accessibility reviews" **kept from the old README**
Python 3.12, Node 22+	`ci.yml` (3.12, 22), Render runtime	True; "Next.js needs Node 20.9+" **kept from the old README**

Claim in the README	Source	Result
Local run steps	Run in the worktree (section 6)	**All work**
`.env` not loaded automatically	`main.py` reads `os.getenv` only; no dotenv	True; the link to `.env.example` **kept from the old README**
Test commands, PostgreSQL migration tests	`ci.yml`, `test_migrations.py`	True
CI	`.github/workflows/ci.yml`: two jobs	**Made exact**: backend tests with a PostgreSQL 16 service; frontend build then Chromium tests
Migrations applied at startup, additive, with a test	`migrate.py`, `DEPLOYMENT.md`	True; the `alembic revision` command **kept from the old README**
Project map	`ls` of each path	True
How it is built (stages, handoffs, decision log)	`build.md` section 14, `docs/handoffs/`, `decision-log.md`	True
Deploying: two web services and PostgreSQL in `render.yaml`	`render.yaml`	True; "the repository is private" **kept from the old README**
"No licence has been chosen yet; all rights are reserved by the owner"	No `LICENSE` file; the repository is private	**Kept, not invented; you must confirm** (question 2)

4. What the draft dropped, and where it went

Every fact from the old README that is still true is now in the README (mostly the new collapsible **More details**) or in `DEPLOYMENT.md`:

Old README fact	Now in
Authlib; signed `HttpOnly` sessions	README, Safety
Invitations: hashed, single-use, email-bound, seven days, shared by the parent, no email sent	README, More details
Family records, account history, practice history	README, More details
Only the active card is shown; illustrated scenes; reduced motion; completions saved, preview in memory	README, More details

Old README fact	Now in
Unit 1's bounded checker (structure only, different names and both paths, any wording, taught syntax only), not a trusted grade	README, More details
15-second stop, output limit, worker CSP limited to the Pyodide CDN, security review before broad registration	README, table and Safety
Accessibility features; WCAG 2.2 AA not audited	README, More details
Node 20.9+; `.env.example`; `alembic revision`	README, Run it locally
Curriculum tests run only trusted authored solutions; live Google sign-in and Render configured	README, Tests (the second is also in `DEPLOYMENT.md`)
Before broader release: retention and deletion, monitoring, backup and restore drills, privacy notice, security and accessibility reviews	README, Known limits
Review addresses: `/courses`, `/courses/<id>`, `/learn/<lesson>`, `/preview?lesson=hello&card=1`, Coming soon not startable, non-students preview	README, More details
Goals: asked once, skipping means "Learn to code from scratch", changeable, only changes the recommendation, released courses stay open, parents see it	README, More details
`/api/check` public and unsaved; `/api/attempts` and `/api/progress` need a student session	README, More details
Lesson sequence and the two Python tutorial links	README, More details
Player replies typed before a run; prompts not in the output	README, More details
Loop practice replaced; records kept; badges mark completed practice	README, More details
Private repository; Blueprint with Starter and basic-256mb and checking the price; `APP_ORIGIN` and `API_ORIGIN` with no trailing slash; `BACKEND_HOSTPORT` automatic, `BACKEND_URL` elsewhere and resolved at build time; Google sign-in off until the URL exists; `SESSION_SECRET` generated; database from the Blueprint; the pre-launch verification list	`DEPLOYMENT.md`, new "Setting up from scratch" (only what it did not already say)
The local Google redirect `http://localhost:3000/auth/google/callback`; never a birthday scope	`DEPLOYMENT.md`, same section

Dropped, because no longer true: - "Add rate limiting/invitation quotas": rate limits, including invitations, have existed since stage 6. - "No Google Cloud project or OAuth credential has been created by this implementation session": the client was created later, as `DEPLOYMENT.md` records.

No secret or ID was added: every ID is already in `DEPLOYMENT.md`.

5. Decisions and why

Decision	Options considered	Reason	Who decided
Commit the draft unchanged first, then the corrections	One commit	Shows exactly what was changed and why, and keeps your draft safe in the repository's history	Implementer
Old facts in a new collapsible **More details** section	Spread them through the visible text; drop them	Keeps the draft's short visual front page while losing nothing; matches its other collapsible sections	Implementer (**assumption 1**)
Result labels as the page shows them	The draft's descriptive words	A reader can match them to what they see	Implementer
Screenshots from `/preview` with the repository's own test data and a mocked API	A real account	No account, email or learner data in any image	Task rule
`catalog.png` cropped just below the second row of courses	The full page	A half-cut third row looked broken	Implementer
`friendly-error.png` and `hint-ladder.png` show the whole lesson card, 896 px wide	Only the editor panel	The task asked for about 900 wide; the card shows the instructions next to the help	Task. **Changed in the polish:** now the editor panel only, at 2× (section 12)
The banner left exactly as drawn	Add `width`/`height` attributes	It has no technical problem where it is used (section 6)	Task rule. **Changed in the polish:** `width` and `height` added at the owner's request; the design is unchanged (section 12)

6. How it was tested

Check	Method	Result
Links, images and anchors	A script reads every Markdown link and every `href` and `src` in `README.md` and `DEPLOYMENT.md`, and checks files and GitHub heading anchors	**15 checked, 0 problems**
External links	`curl`	The live site, the preview and both Python tutorial pages: 200

Check	Method	Result
Mermaid diagram	`mmdc` (`@mermaid-js/mermaid-cli` 11.4.2 through `npx` in a scratch folder, with Playwright's Chromium)	Renders: browser, worker, frontend, backend, PostgreSQL, Google
GitHub rendering	`gh api markdown` (GitHub's own renderer, `gfm` mode), viewed with GitHub's stylesheet at 1012 px and 400 px	Kept: the centered header, the link row, all four tables, both image widths, the side-by-side cells and the three collapsible sections. No broken images, no sideways scrolling. Findings in section 7
Banner	`xmllint`; Chromium alone and inside an `<img width="100%">`	Valid; 4,623 bytes; no scripts or outside links; renders as drawn; scales to 900 × 267 in the README's `<img>`
Screenshots	Viewed each one; sizes	`catalog.png` 1280 × 864, 161 KB; `friendly-error.png` 896 × 1327, 125 KB; `hint-ladder.png` 896 × 1381, 147 KB. All under 300 KB
Local run steps	Fresh Python 3.12 virtual environment, `pip install -r requirements.txt` (with `uv pip`), `uvicorn` with `APP_ORIGIN` and `API_ORIGIN` set, `npm ci`, `npm run dev`	API `/health` 200, log "Database schema is at migration 0008"; `/`, `/courses`, `/preview?lesson=hello` and `/health` 200 through the dev server; `/api/lessons` 200 and `/api/me` 401 through the proxy. Google sign-in was not tried: there is no local OAuth client, as the README says
Nothing else changed	Full suites in the worktree	Backend **358 passed, 11 skipped** (SQLite), **369 passed** (PostgreSQL); build OK; Playwright **108 passed**

Deliberate-break controls: none, because there is no code to break. The link script is the automated check.

Bugs found: none in code. In the draft, the wrong things were wording, not facts that could hurt anyone: the result labels, one course title, and missing numbers. Near-miss in my own preview: a "Loading" spinner appeared under the diagram, because I swapped GitHub's diagram block for the rendered picture but left its placeholder. It does not appear on GitHub.

7. How it renders, and what renders less well

> The first three points below were fixed in the polish (section 12). The "What is coming" wrap is unchanged.

- **Fine:** the banner, the centered link row and catalog image, the collapsible sections, the tables, and the diagram (GitHub draws it on the page).
- **The feature table's empty header row** (`| | |`) shows on GitHub as a thin empty bar above the table. Not broken, but visible.

- **The side-by-side screenshots are small.** At GitHub's width each is about 470 px, so the text inside them is hard to read; on a phone it is unreadable. Options: crop each to the editor panel (about 450 px wide, so the text shows larger), or stack them one under the other. I kept the layout you asked for.
- **"What is coming"**: the Status column is narrow, so "Unit 1 released; units 2 to 12 planned" wraps over three lines.
- **The banner in other viewers:** it has no `width` or `height` attributes, so a viewer that ignores the README's `width="100%"` would show it at 300 px. On GitHub this does not happen. Adding `width="1280" height="380"` to the `<svg>` tag would fix it everywhere, without changing the design. Tell me if you want that.

8. Questions for the owner

1. **"It started as a way to help one grade 9 student…"**: this is not written anywhere in the repository. Is it right as worded?
2. **Licence:** the README says no licence has been chosen and all rights are reserved. I did not choose or add one. Do you want a licence, or a different sentence (for example "Private repository; not licensed for reuse")?
3. ~~The empty header row, the small screenshots and the banner's size attributes~~: answered; done in the polish (section 12). Still open: the screenshots are side by side on a phone too, so they are small there (section 12). Keep that, or stack them on narrow screens?
4. **Assumption 1:** the old README's facts in a collapsible "More details" section. OK?

9. Where the drafts went

- `README-draft-from-chat.md` and `README-banner-from-chat.svg` are **no longer in the main folder**. They were moved to my session's scratch folder, `/private/tmp/claude-502/-Users-DEV-Projects-python-academy/d91ca5bb-1855-42f9-95a1-82f13fcfcfc9/scratchpad/`. That folder is temporary and may be cleaned up later.
- Both are also preserved unchanged in this PR's first commit, `9692b03`, as `README.md` and `docs/images/banner.svg`. To get the draft back: `git show 9692b03:README.md`.

10. Next concrete action

Answer section 8, review and merge. After PR #27 (UI-05) merges, update "Near-term work" in the README, which still lists lessons per unit as coming.

11. Removing the worktree safely

Everything in `/Users/DEV/Projects/python-academy-readme` is committed and pushed. Only ignored files (`node_modules`, `.next`, caches) and the untracked `frontend/AGENTS.md`, which Next.js creates when it

builds, are left. Because that file is untracked, `git worktree remove` refuses unless it is gone first. From the **main** folder:

```
cd /Users/DEV/Projects/python-academy
git -C ../python-academy-readme status --short   # shows only ?? frontend/AGENTS.md
rm ../python-academy-readme/frontend/AGENTS.md   # generated by Next.js, not part of the
project
git worktree remove ../python-academy-readme     # deletes only that folder; refuses if
anything else is uncommitted
git worktree prune
```

After this PR is merged, `git branch -d docs/readme` deletes the local branch; `-d` refuses if it is not merged. Don't delete the folder in the Finder or with `rm -rf`: Git would keep a stale record of it (`git worktree prune` fixes that). The UI-05 worktree (`../python-academy-ui05`) and the FB-04 worktree (`../python-academy-fb04`) are removed the same way, after `rm`-ing their own `frontend/AGENTS.md` if `git status` shows one.

12. Polish (after the owner's review)

Same branch and worktree, documentation only.

Change	What was done	Checked
Screenshots cropped to the code panel	`friendly-error.png` and `hint-ladder.png` were taken again at 2× pixel density (same `/preview` page, same test data, mocked API) and cut to the right-hand panel only: from the "story.py / Python playground" header down to the last button above the "Python runs in your browser…" note. The crop sits 6 px inside the card's edge so its rounded corner stays out. Now **870 × 2282, 139 KB** and **870 × 2391, 181 KB** (435 CSS px wide). In the README each `<img>` has `width="380"` and its cell `valign="top"`. The alt text now describes only what each crop shows	Viewed both images; under 300 KB
Feature table header	`| | |` replaced by real headers, **What** and **What it does**	GitHub render: two header cells with text, no empty bar
Banner size	`width="1280" height="380"` added to the `<svg>` tag. Nothing else in the file changed (one-line diff)	Renders the same inside the README's `<img width="100%">`

How it was checked: the link script again (**15 checked, 0 problems**); the README through `gh api markdown` again, viewed with GitHub's stylesheet at 1012 px, 830 px and 400 px wide:

Width	Screenshots	Feature table	Banner	Sideways scroll
1012 px (desktop)	Side by side, 380 px each, text readable	Headers "What" / "What it does"	948 × 281	None
830 px (narrow desktop)	Side by side, 356 px each	Same	766 × 227	None
400 px (phone)	Side by side, **141 px each: too small to read**; tapping one opens it full size on GitHub	Same	336 × 100	None

Suites, unchanged since nothing outside docs changed: backend **358 passed, 11 skipped** (SQLite), Playwright **108 passed**. No deliberate-break check: there is no code to break.

Still rendering less well: on a phone the two screenshots stay side by side, so each is small. Stacking them would need a different layout than the side-by-side one you asked for (question 3). The "What is coming" Status column still wraps.